

Quadratic Simultaneous Equations

AQA Higher · GCSE Grade 8–9

Name: _____ Class: _____ Date: _____

What you need to be able to do

Solve a pair of equations where one is a straight line (linear) and one is a curve (quadratic), using substitution. Most pairs have two answer points — find both x values and the y that goes with each.

Common mistakes to avoid

- There are usually two answers, not one. Find both x values — for example $x = 3$ and $x = -1$.
- Get everything on one side so it equals 0 first. For example: $x^2 = 2x + 3$ becomes $x^2 - 2x - 3 = 0$.
- Watch the minus signs. $(x - 1)^2 = x^2 - 2x + 1$, not $x^2 - 1$.
- Each x has its own y. After you find x, put it back in to work out y.

Method steps

1. Step 1: Make y (or x) the subject of the straight-line equation.
2. Step 2: Substitute that into the quadratic equation.
3. Step 3: Move everything to one side so it equals 0.
4. Step 4: Solve the quadratic — factorise or use the formula.
5. Step 5: Put each value back to find its pair. Write answers as (x, y).

Worked example

Solve: $y = x^2 - 2x + 3$ and $y = x + 3$

Substitute: $x^2 - 2x + 3 = x + 3$

Rearrange: $x^2 - 3x = 0$

Factorise: $x(x - 3) = 0 \rightarrow x = 0$ or $x = 3$

Find y: $x = 0 \rightarrow y = 3$; $x = 3 \rightarrow y = 6$

Answers: (0, 3) and (3, 6)

Question 1 of 8

[3 marks] ●○○○○

Solve the simultaneous equations.

$$y = x^2 \text{ and } y = 2x + 3$$

How to start

1. First, both equations already give y , so set them equal: $x^2 = 2x + 3$
2. Next, move everything to one side so it equals 0: $x^2 - 2x - 3 = 0$
3. Then factorise and solve to find the two x values.
4. Finally, put each x back into $y = 2x + 3$ to find its y .

$x =$ _____

$y =$ _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Question 2 of 8

[3 marks] ●○○○○

Solve the simultaneous equations.

$$y = x^2 + 1 \text{ and } y = x + 3$$

How to start

1. Set the two right-hand sides equal, then rearrange to $= 0$.
2. Factorise to find the two x values, then find each matching y .

x = _____

y = _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it $= 0$
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Question 3 of 8

[3 marks] ●●○○○

Solve the simultaneous equations.

$$y = x^2 - 2x \text{ and } y = x + 4$$

x = _____

y = _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Question 4 of 8

[3 marks] ●●○○○

Solve the simultaneous equations.

$$y = x^2 - 5x + 6 \text{ and } y = x - 2$$

x = _____

y = _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Question 5 of 8

[4 marks] ●●●○○

Solve the simultaneous equations.

$$x^2 + y^2 = 25 \text{ and } y = x - 1$$

x = _____

y = _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Question 6 of 8

[4 marks] ●●●○○

Solve the simultaneous equations.

$$x^2 + y^2 = 10 \text{ and } x + y = 4$$

x = _____

y = _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Question 7 of 8

[5 marks] ●●●●○

Solve the simultaneous equations. Give your answers in surd form.

$$x^2 + y^2 = 9 \text{ and } y = x + 1$$

x = _____

y = _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Question 8 of 8

[5 marks] ●●●●●

A rectangle has an area of 12 cm^2 and a perimeter of 14 cm .

By forming and solving a pair of simultaneous equations, find the length and width of the rectangle.

(Let the length be x and the width be y .)

$x =$ _____

$y =$ _____

Remember the method:

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

Which question did you find the hardest, and why?

One method step I want to remember next time:

One thing I will practise before the next lesson:

Teacher key

Answers

1. Q1: (3, 9) and (-1, 1)
2. Q2: (2, 5) and (-1, 2)
3. Q3: (4, 8) and (-1, 3)
4. Q4: (4, 2) and (2, 0)
5. Q5: (4, 3) and (-3, -4)
6. Q6: (1, 3) and (3, 1)
7. Q7: $x = \frac{-1 + \sqrt{17}}{2}$, $y = \frac{1 + \sqrt{17}}{2}$ and $x = \frac{-1 - \sqrt{17}}{2}$, $y = \frac{1 - \sqrt{17}}{2}$
8. Q8: length 4 cm, width 3 cm (from $xy = 12$ and $x + y = 7$)